



# Influences on pre-service teachers' intention to use classroom management strategies for students with ADHD: A model analysis<sup>☆</sup>



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## ABSTRACT

Students with ADHD often feel mistreated at school, receive low grades, and can have low socio-economic statuses as adults. Evidenced-based classroom management strategies (CMS) are effective in minimizing ADHD-related issues in schools, but have not found their way into practice. 1086 pre-service teachers completed an online survey on direct experiences, social influences, individual differences, attitude, and intention to use CMS. We examined which variables mostly influenced the intention to use CMS. Pre-service teachers' attitude towards CMS was the most important influencing factor, and knowledge most significantly influenced attitudes towards CMS. Attitude towards CMS need to be changed to increase the probability pre-service teachers will use effective CMS. Additionally, education for teachers should include information on CMS.

## 1. Introduction

"The vice-principal even said straight to me 'You look like that possessed kid from *The Exorcist*.'" This is a quote from an interview with a child with attention deficit/hyperactivity disorder (ADHD) (Honkasilta, Vehkakoski, & Vehmas, 2016), impressively illustrating the difficult relationship students with ADHD often have with their teachers, as well as the negative generalized expectations they often face. Students with ADHD often feel misunderstood and treated unfairly, and their teachers feel overwhelmed by increasing classroom stressors and a lack of knowledge needed to deal with students' behavior (Greene, Beszterczey, Katzenstein, Park, & Goring, 2016; Honkasilta et al., 2016). Children with ADHD also have positive experiences, such as supporting teacher behavior (Honkasilta et al., 2016). This implies there are individual differences in how teachers interact with students with ADHD, which can be linked to their use of classroom management strategies (CMS) (Clunies-Ross, Little, & Kienhuis, 2008). Generally, effective and ineffective CMS can be distinguished. Effective ones are antecedent - based, (e.g. setting classroom rules, modifying tasks) and consequent - based, (e.g. using a token plan or use reprimands, or self-management strategies, with which the student is able to monitor its own behavior) (Gaastra, Groen, Tucha, & Tucha, 2016). In a German survey only about 48 % of the teachers' mentioned strategies to handle ADHD in the classroom were classified as effective ones, though (Ruhmland & Christiansen, 2017). Ineffective CMS are by far used more often and based on coercing and punishment (e.g. shouting at students with ADHD, sending the student out of the room with no comment, or telling the student that he/she should behave or concentrate better) (Hoberg, 2013). Although impressive evidence supports positive CMS as effective in preventing and addressing students' symptoms of ADHD, including disruptive behavior, in the classroom (Barkley et al., 2000; Christiansen, Hirsch, König, Steinmayr, & Roehle, 2015; Gaastra et al., 2016;

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Miranda, Presentación, & Soriano, 2016), their use in practice is rare (Ruhmland & Christiansen, 2017). Instead, ineffective CMS are reported to be used by far more often (Ruhmland & Christiansen, 2017). This implies a science-practitioner gap where the current state of research does not find its way into the classroom. Hence, the aim of the present study is to detect relevant factors impacting teachers' intentions to use CMS to help work towards closing this gap. Our second aim is to identify factors influencing teachers' intentions to use ineffective CMS in order to reduce the usage of ineffective CMS. We conducted our study based on a large sample of pre-service teachers.

### 1.1. Expectations and attitudes

Teachers often have generalized expectations regarding ADHD, which can influence their behavior towards students diagnosed with the disorder (Ohan, Visser, Strain, & Allen, 2011). Expectations, as defined by Rief et al. (2015) in the *violation-expectation model* (ViolEx) propose "if-X-then-Y" sequences that emerge from learning processes, in which a unique situation, "X," (e.g., using CMS for a misbehaving student) leads to the anticipation of a unique response "Y," (e.g., CMS will not work and student will be strenuous). If "Y" does not happen (CMS works and student behaves positively), expectations are violated. This might result in three different general responses: 1) accommodation, in which case learning takes place and students with ADHD might be viewed more differentially; 2) assimilation and 3) immunization - in both cases no learning takes place, as the unexpected information will be discarded (Rief et al., 2015). In case of assimilation individuals actively remove any discrepancies between their generalized expectations and expectation inconsistent outcomes without changing their expectations. For instance, people avoid expectation inconsistent outcomes in the future and/or contribute actively to a higher likelihood of expectation consistent outcomes in the future (i.e., "self-fulfilling prophecy," Miller & Turnbull, 1986; Stinson, Logel, Shepherd, & Zanna, 2011, "active inference," Friston, 2009; Hechler, Endres, & Thorwart, 2016). In case of immunization individuals minimize the potential impact of discrepant information on their expectations (e.g. "that the child behaved positively was an exception"). The latter is confirmed by studies from social psychology demonstrating that confronting people with stereotype inconsistent information does not necessarily change their stereotypes, but leads to the judgement that this information is an atypical fact and can thus be ignored (Carnaghi & Yzerbyt, 2007; Kunda & Oleson, 1995).

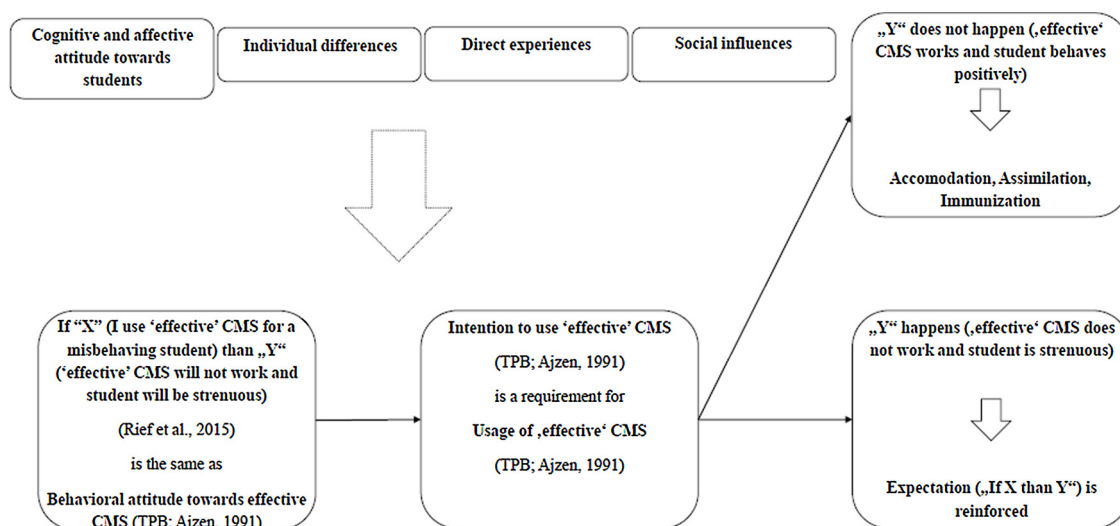
Significant variables, such as direct experiences, social influences, and individual differences, influence the mechanisms of "if X then Y" sequences in the ViolEx model. As we use the ViolEx model as a framework of the described and assumed science-practitioner gap in using evidence-based effective CMS for students with ADHD we consider the variables generalized expectations, direct experiences, social influences, and individual differences.

According to Ajzen (2005), generalized expectations towards CMS can be measured as behavioral attitude and are influencing behavior. For instance, research showed that a teacher's attitude towards inclusion is a relevant factor that changes a teacher's intention to support inclusion (Lübke, Meyer & Christiansen, 2016). Ajzen (2005) determined that *attitude* can be categorized as either attitude towards an object (e. g. students with ADHD) or attitude towards behavior related to an object (e.g. attitude towards CMS' usage). The first consists of expectations regarding cognition and affect towards the attitudes' object and the individual subjective evaluation of those. The second is *behavioral attitude*, and it would most likely influence a teacher's intention to use CMS. It can be measured as the expected probability a behavior will have consequences, multiplied by the subjective evaluation of the consequences (e.g., CMS will be effective in reducing the disruptive behavior of students with ADHD). Therefore, behavioral attitude comprises the "If X than Y" sequences and makes them measurable. Additionally, three other factors may affect a teacher's intention to use CMS: attitude towards students with ADHD, perceived behavioral control, and subjective norms, as defined by the *theory of planned behavior* (TPB; Ajzen, 1991).

According to the TPB, human behavior is influenced directly by the intention to execute a behavior and by perceived behavioral control. Intention itself is, in turn, influenced by perceived behavioral control, subjective norms, and attitudes towards a behavior (Ajzen, 1991). *Intention* is the motivation behind a behavior. The teachers' *behavior* in our understanding would be to implement an effective/ineffective CMS. *Perceived behavioral control* is one's ability to handle a specific situation (e.g. I will be able to handle a strenuous student). The *subjective norm* is one's motivation to meet the expectations of a relevant group regarding a behavior (e.g. the headmaster expects me to handle my classroom well) (Ajzen, 1991, 2005). In the present study, we focused on the influence of intentions to identify the most important variables affecting teachers' intentions to use CMS. In Fig. 1 the theory of planned behavior (Ajzen, 1991) and the violation expectation model (Rief et al., 2015) and its usability to assess teachers' intention to use CMS are summed up.

### 1.2. Direct experiences

Generalized expectations and therefore also the attitude are largely influenced by direct experiences (Rief et al., 2015). ADHD-related behavior in classrooms, such as inattention, impulsivity, and hyperactivity, can be difficult to handle. Compared to other students, teachers can feel significantly more stressed by students with ADHD (Greene et al., 2016). Furthermore, pre-service teachers that are defined in our study as education majors, that are not or not regularly teaching, generally show a more favorable response to students with ADHD than experienced in-service teachers (Anderson, Watt, Noble, & Shanley, 2012). Hence, teachers may often have negative experiences with, as well as stress from students with ADHD, leading teachers to expect further negative interactions with these students. Thus, it is important to understand how much of teachers' stress is due to the classroom behavior of students with ADHD, as this may negatively influence teachers' attitudes towards, and intentions to use, CMS.



**Fig. 1.** Expectations (“If X than Y”) influence the usage of classroom management strategies (CMS). Because of individual differences, direct experiences, social influences and the attitude towards students with ADHD a specific expectation (If X [I use ‘effective’ CMS for a misbehaving student] than Y [‘effective’ CMS will not work and student will be strenuous]) is built. In a specific situation the expectation is activated and the intention to use effective CMS is minimized. However, if it did come to a usage of effective CMS two things could happen. If the assumption “Y” (‘effective’ CMS does not work and student is strenuous) holds true, the expectation (“If X than Y”) is reinforced. If “Y” does not hold true and the ‘effective’ CMS works and students behaves positively three possible mechanisms (accommodation, assimilation, immunization) might happen and it comes either to a change in the “If X than Y” sequences or not.

### 1.3. Social influences

Social experiences shape expectations and therefore also the attitude through contact and communication with peer groups about the sometimes-implicit knowledge of people around them (Rief et al., 2015). From this knowledge, individuals often act in a way that generalized expectations create a social reality and thus self-confirm (Jussim, 2012; Rief et al., 2015). Prejudices, stereotypes, and widespread opinions about other groups are the expression of generalized expectations and an important factor in the decision process (Rief et al., 2015). Previous findings showed that teachers’ affects towards students with ADHD are often negative (Anderson et al., 2012; Mulholland, Cumming, & Jung, 2015), and students with a diagnosis of ADHD were rated as worse academically and more impaired than comparable children demonstrating the same behavior, but without the diagnostic label of ADHD (Ohan et al., 2011). Furthermore, teachers have described reduced confidence in their abilities to handle students with ADHD (Ohan et al., 2011). Subjective norm in the context of the TPB measures teachers’ socially shared knowledge but also their willingness to meet social norms (Ajzen, 2005).

### 1.4. Individual differences

Frequently experiencing stress can lead to increased mental health issues for teachers, as evidenced by research showing teachers have a significantly increased risk of developing somatization disorder (Howard et al., 2017) or depressive symptoms (Hammen & DeMayo, 1982). Feelings of stress and psychological strain teachers experienced directly were linked to declines in classroom engagement and teacher-student relationships (Hakanen, Bakker, & Schaufeli, 2006; Yoon, 2002).

Perceived behavioral control, as mentioned in the TPB influences not only the intention to execute a behavior, but also the attitude towards a behavior (Ajzen, 1991).

Research shows expectations and therefore also the attitude are influenced by different personality traits, and in the TPB, Ajzen and Fishbein (2005) described personality factors as influencing variables of attitude. Ekehammar, Akrami, Gylje, and Zakrisson (2004) showed that *right-wing-authoritarianism* (RWA), *social dominance orientation* (SDO), and the big five personality traits matter most in predicting prejudices. Therefore, we focused on those variables. The concept of “the authoritarian personality,” was constructed to examine why some people have prejudices towards minorities rather than others (Adorno, Frenkel-Brunswick, Levinson, & Sanford, 1950). This concept was later revised to be RWA and measures a liableness to fascist ideologies (Altemeyer, 1981, 1998). SDO was first introduced by Pratto, Sidanius, Stallworth, and Malle (1994), and refers to a perspective in which one has a preference for hierarchical societal structures over more equal ones.

The “Big Five” refers to five personality dimensions (*openness, conscientiousness, extraversion, agreeableness, neuroticism*) that describe differences in personality cross-culturally, and have a shaping function regarding expectations and therefore also the attitude (DeYoung, 2015; Ekehammar et al., 2004).

*Stress Reactivity* represents our vulnerability in reacting to stressful events. Stress reactivity can explain the varying stress levels of different people in same situations (Schulz, Jansen, & Schlotz, 2005). People with a high level of stress reactivity have inter alia a

decreased concept of their abilities and difficulties in setting new goals for themselves (Schulz et al., 2005), which directly influences their expectations and therefore also the attitude towards CMS.

As part of the individual differences outlined above, teachers' knowledge about ADHD has been identified as an additional factor influencing their expectations and therefore their attitude (Bekle, 2004). Several studies demonstrated that teachers often do not have accurate knowledge on ADHD, and this is linked to more coercive teaching behavior, which can aggravate students' behavior problems (Ruhmland & Christiansen, 2017; Schmiedeler, 2013; Soroa, Gorostiaga, & Balluerka, 2016). Teachers' lack of knowledge about ADHD is especially pronounced regarding treatment opportunities and the prevalence of the disorder (Bekle, 2004; Kos, Richdale, & Hay, 2006; Sciuotto, Terjesen, & Frank, 2000).

### 1.5. Research questions

In this study, we explore pre-service teachers' behavioral attitudes as a proxy for their generalized expectations. As these influence the intention to either use effective or ineffective CMS, we focus on factors that were identified as relevant: direct experiences, social influences, and individual differences. To improve our understanding of the science-practitioner gap, including its potential causes and why some teachers may cope better with students with ADHD than others, we aimed to determine what variables shape, and to what extent, both attitudes towards, and intentions to use, CMS. Knowledge on influencing variables shaping pre-service teachers' intention to use effective or ineffective CMS lead to implications for teacher education as well for further research. Implementing more effective and less ineffective CMS for students with ADHD leads to reduced impairment of children with ADHD, a higher equality of opportunities for students with ADHD and makes best use of teachers' potential.

The following research questions are explored:

- 1 Can the intention to use effective CMS be explained by pre-service teachers' behavioral attitude towards effective CMS as well as through individual differences, direct experiences, social influences, and their attitude towards students with ADHD?
- 2 Are all variables that influence behavioral attitude towards effective CMS also influencing the intention to use effective CMS mediated through behavioral attitude towards effective CMS?
- 3 Can the intention to use ineffective CMS be explained by pre-service teachers' behavioral attitude towards ineffective CMS as well as through individual differences, direct experiences, social influences, and their attitude towards students with ADHD?
- 4 Are all variables that influence behavioral attitude towards ineffective CMS also influencing the intention to use ineffective CMS mediated through behavioral attitude towards ineffective CMS?

## 2. Methods

### 2.1. Sample population

Using the online tool SoSciSurvey (<https://www.soscisurvey.de/>), we conducted the present study with pre-service teachers from all over Germany. The sample population included 1086 participants ( $n = 1086$ ). A total of 1741 pre-service teachers ( $n = 1741$ ) started the survey, and 1118 completed it ( $n = 1118$ ). Consequently, the dropout rate for the survey was 35.78 %. Furthermore, 2.86 % ( $n = 32$ ) of the participants were excluded from the analysis, because they completed less than 75 % of at least one scale. The mean age of the participants was 23.22 years ( $SD = 3.93$ ), and 30.57 % were male, 68.97 % were female, and 0.46 % identified as another gender, not further specified. Table 1 lists the participants' schools they were prepared to teach for. Predominantly, academic high school pre-service teachers participated in the survey.

### 2.2. Procedure and sequence of the survey

The study adopted a cross-sectional, non-experimental design to test the suggested variables. We sent the link leading to the survey via e-mail to universities with the request it be passed on to their students. Furthermore, the link was distributed in Facebook groups for pre-service teachers. The subject lines of the e-mail and the Facebook posts included information regarding possible prizes for the participation, to be awarded via raffle. It was possible to win a Nintendo Switch (approximate value 300 euros) or one of three

**Table 1**

Pre-service teachers' current course of study. The percentages can be more than 100, because more than one program can be studied at a time.

| School type                         | %     | Number (absolute) |
|-------------------------------------|-------|-------------------|
| Primary school (Grundschule)        | 27.72 | 301               |
| Comprehensive school (Gesamtschule) | 7.32  | 80                |
| Main school (Hauptschule)           | 9.12  | 99                |
| Middle school (Realschule)          | 16.30 | 177               |
| Academic high school (Gymnasium)    | 45.03 | 489               |
| Special needs school (Förderschule) | 10.04 | 109               |
| Other (Sonstige)                    | 2.39  | 26                |

gift certificates, with a value of 50 euros each. The survey included detailed information on the purpose of the study, which was also described briefly in the e-mail and the Facebook posts. The survey started in November 2017, and was available for five weeks, with a reminder sent to the universities after three weeks. After clicking on the link, all participants received detailed study information and were asked to provide informed consent. This was followed by questions on participants' age, gender, and course of study, and an open-ended question on their estimation regarding the average number of students with ADHD per classroom. At the end of the survey, all participants were asked whether they wanted to receive further information on the study and if they would like to take part in the raffle. If they decided to choose one or both of those options, they were asked to provide their e-mail addresses, which were saved separately from the survey data.

### 2.3. Measures

#### 2.3.1. Intention to use CMS

To measure the intended use of effective and ineffective CMS the intervention section of the ADHD School Expectation Questionnaire (ASE) was used (Dort et al., under revision). The ASE consists of a knowledge, an intervention, and an attitude scale and was validated with pre-service and in-service teachers. Construct validity and reliability are overall satisfactory (Dort et al., under revision). The intervention scale consists of 15 effective and twelve ineffective interventions, which were selected by reviewing previous findings (Dupaul, Eckert, & Vilardo, 2012; Miranda et al., 2016). The respondent is required to estimate how often he/she would use the intervention. For example, items of effective CMS are "Use a reward system" or "Behavior related praise as often as possible". Items representing ineffective CMS are "Tell the student to behave better" or "Banish the student from the classroom with no comment". The assessment of the ASE is based on a twelve-point Visual Analogue Scale (VAS) ranging from *never* to *very often*. It was transformed afterwards into an interval scale from zero to one. The scale was divided into subscales for either effective (15 items) or ineffective (twelve items) CMS. Cronbach's  $\alpha = .80$  for the effective CMS scale ( $M = .72$ ,  $SD = .12$ ),  $\alpha = .79$  ( $M = .35$ ,  $SD = .14$ ) for the ineffective CMS scale, and  $\alpha = .73$  ( $M = .55$ ,  $SD = .09$ ) for the total scale.

#### 2.3.2. Generalized expectations: Attitude with affective, cognitive, and behavioral components

Affective and cognitive attitudes towards students with ADHD were measured using the ASE (Dort et al., under revision). It comprises thirty-three items on attitudes towards students with ADHD. The scale measuring the cognitive attitude component addresses behavior that is expected of students with ADHD (27 items) and the rater indicates whether this behavior is *likely* or *unlikely* (coded from 0 to 1) as well as *negative* or *positive* (coded from -3 to +3). The two ratings were multiplied as two components and added together afterwards constituting one joint measurement of attitude, in reference to Ajzen (1991). Sample items are "Loses school material", "Can concentrate", "Avoids tasks that demand attention." Affective attitude was equally measured by requesting for ratings of potential emotions a (pre-service) teacher may feel towards students with ADHD (six items). The queried emotions were among others "joy" and "anger." Cronbach's  $\alpha = .87$  ( $M = -11.66$ ,  $SD = 13.92$ ) for the attitude scale.

The behavioral component of the attitude scale comprises twenty-seven CMS that can be used in the classroom to reduce disruptive behavior (e.g. "Short, concise instructions."), and is rated on a VAS ranging between *not effective at all* and *very effective*. For the effective CMS' subscale Cronbach's  $\alpha = .82$  ( $M = .73$ ,  $SD = .13$ ), for the ineffective CMS' subscale Cronbach's  $\alpha = .74$  ( $M = .52$ ,  $SD = .09$ ), and for the total scale Cronbach's  $\alpha = .73$  ( $M = .26$ ,  $SD = .12$  scale).

#### 2.3.3. Direct experiences

2.3.3.1. *Stress*. One item ("How high do you estimate your stress will be from the behavior of students with ADHD in the classroom?") assessed expected individual stress levels caused by students with ADHD. This was rated on a VAS ranging from *not at all* to *very massive*. The mean of the item was  $M = 3.18$ , with answers ranging from zero to five, with a standard deviation of  $SD = 0.98$ .

#### 2.3.4. Social influences

2.3.4.1. *Subjective norms*. Six items on subjective norms were constructed based on the TPB (2002b, Ajzen, 2002a) assessed with a VAS ranging from *I totally disagree* to *I totally agree*, that was transformed to range from zero to five. Those six items resulted in an unsatisfactory Cronbach's  $\alpha$  of .32 ( $M = 2.90$ ,  $SD = 0.63$ ). We shortened the scale to improve Cronbach's  $\alpha$  (e.g. items with discriminatory power below .30 were eliminated; Blanz, 2015); this resulted in  $\alpha = .52$  ( $M = 3.12$ ,  $SD = .98$ ). The items are "I want people who are important to me to think well of me", "Doing something that I know others consider as unethical, makes me lose my self-esteem", and "I do not care if other people have a bad opinion on me."

#### 2.3.5. Individual differences

2.3.5.1. *Psychological strain*. Psychological strain was measured using the Brief Symptom Inventory (BSI) (Franke, 2000). The BSI consists of fifty-three items grouped into nine scales, which form three global characteristics: Global Severity Index (GSI), Positive Symptom Distress Index, and Positive Symptom Total. GSI measures mental strain in general, and this index was used to identify the level of psychological stress for pre-service teachers. A sample item is "How much have you suffered from nervousness or tremors in the past seven days?" A four-point Likert-type scale ranging from *nothing at all* to *very strong* was used. Franke (2000) reported a Cronbach's  $\alpha = .95$  for the GSI, and in the present study it was  $\alpha = .91$  ( $M = 1.25$ ;  $SD = 0.2$ ).

2.3.5.2. *Perceived behavioral control*. Two items were generated to measure perceived behavioral control. Item compilation was based on the specifications of the TPB (Ajzen, 2002a, 2002b). The items are "I have the ability to teach students with ADHD effectively" and

“Dealing with students with ADHD exceeds my ability”. Comprehensibility was tested in a pre-test. The answering format was a VAS varying from *I totally disagree* to *I totally agree* and ranging from zero to five. Cronbach’s  $\alpha = .84$  ( $M = 2.6$ ,  $SD = 1.28$ ).

**2.3.5.3. Right-wing authoritarianism (RWA).** The German short scale on right-wing authoritarianism (KSA-3) by [Beierlein, Asbrock, Kauff, and Schmidt \(2014\)](#) was used to measure RWA, with the three associated subscales (three items per subscale; e.g. “*Society should be tough on outsiders and idlers.*”). All subscales are measured using a six-point Likert-type scale ranging from *I totally disagree* to *I totally agree*. In the present study, the overall Cronbach’s  $\alpha = .82$  ( $M = 1.90$ ;  $SD = .80$ ), and varied between  $\alpha = .68$  and  $\alpha = .69$  for the different subscales. Thus, we did not use the subscales, but the aggregate mean value of RWA.

**2.3.5.4. Social dominance orientation (SDO).** SDO was measured using the 12-item German SDO scale designed by [Cohrs, Moschner, Maes, and Kielmann \(2005\)](#), in reference to the scale by [Pratto et al. \(1994\)](#) and the German adaption by [Six, Wolfrath, and Zick \(2001\)](#). The answer format is a six-point Likert-type scale ranging from *I totally disagree* to *I totally agree* (e.g. “*We need strong leaders so that we can live safely in society.*”). Cronbach’s  $\alpha = .81$  ( $M = 1.12$ ,  $SD = .65$ ) in the present study.

**2.3.5.5. Big five personality traits.** The German translation of the Big Five Inventory (BFI) ([Rammstedt & Danner, 2016](#)) measures economically five personality factors. All facets are measured using a five-point Likert-type scale ranging from *very inaccurate* to *very applicable* (e.g. “*I do tasks thoroughly.*”). In our sample, the following Cronbach’s alpha coefficients were obtained: Extraversion  $\alpha = .86$  ( $M = 3.65$ ;  $SD = .63$ ), Agreeableness  $\alpha = .78$  ( $M = 3.83$ ;  $SD = .5$ ), Conscientiousness  $\alpha = .85$  ( $M = 3.71$ ;  $SD = .6$ ), Neuroticism  $\alpha = .82$  ( $M = 2.85$ ;  $SD = .64$ ), and Openness  $\alpha = .77$  ( $M = 3.61$ ;  $SD = .6$ ).

**2.3.5.6. Stress reactivity.** To measure stress reactivity, the Perceived Stress Reactivity Scale (PSRS) was used ([Schlotz, Yim, Zoccola, Jansen, & Schulz, 2011](#)). It is a 23-item questionnaire with five subscales and one total scale (four to five items per subscale; e.g. “*When tasks and duties build up to the extent that they are hard to manage*” with the answer possibilities: “*I am generally untroubled, I usually feel a little uneasy, I normally get quite nervous.*”). The authors validated the questionnaire using a three-step answer format, each with a visual anchor, which we also used. In our study we only used the total score, and Cronbach’s  $\alpha = .86$  ( $M = 20.74$ ,  $SD = 7.40$ ).

**2.3.5.7. Knowledge.** The knowledge scale used in the present study includes twenty-four items on ADHD-specific knowledge and is part of the ASE ([Dort et al., under revision](#)). Every correct answer is worth one point and every incorrect answer is worth zero points, with 24 possible points in total. Items were designed to cover the areas etiology, diagnostic & prevalence and intervention, exemplarily items are “*For the clarification of ADHD many diagnostic sources are necessary*”, “*False parenting causes ADHD*”, and “*One single appointment at a psychologist/doctor is sufficient to examine the presence of ADHD.*” Cronbach’s  $\alpha = .85$  ( $M = 7.32$ ,  $SD = 4.21$ ) in the present study.

## 2.4. Statistical analyses

All analyses were performed using the statistical package IBM® SPSS®24.0 and Mplus 8.4 ([Muthén & Muthén, 1998-2017](#)). All raw data have been stored at the Department of Clinical Child and Adolescent Psychology at Philipps University in Marburg, Germany. Inter-correlations between the variables were inspected. A correlation of  $|r| = .10$  is interpreted as a low coherence, of  $|r| = .30$  as a moderate coherence, and  $|r| = .50$  as a strong coherence ([Cohen, 1988](#)). Linear path analysis was used to determine the direction of the correlation, and to identify the explained variance. Mediation analysis was used to examine whether the indirect effects between variables were significant. To determine this, bootstrapping with 10,000 iterations was used. Bootstrapping is a statistically powerful method that is independent of the samples’ distribution and not susceptible for the type one standard error ([Hayes, 2009](#); [MacKinnon, Lockwood, & Williams, 2004](#); [Preacher, Rucker, & Hayes, 2007](#)). Path and mediation analysis were performed with Mplus 8.4 ([Muthén & Muthén, 1998-2017](#)), which uses OLS regression yielding standardized path coefficients for direct and indirect effects. The fit of the model examined through the analysis was determined by the following fit indices: At first, the chi-square test of model fit was inspected. A non-significant value ( $p > .05$ ) means that the model is not significantly different from the data ([Geiser, 2010](#)). Secondly, a significant  $p$ -value ( $< .05$ ) examined by the chi-squared test for the baseline model means that the variables are not uncorrelated ([Geiser, 2010](#)). The Comparative-Fit-Index (CFI) and the Tucker-Lewis-Index (TLI) are indicative for a good model fit when they are at least  $> 0.95$  and rather  $> 0.97$  ([Geiser, 2010](#)). The Root-Mean-Square-Error-of-Approximation (RMSEA) and the Standardized-Root-Mean-Square-Residual (SRMR) are indicative for a high model fit if  $< 0.05$  ([Geiser, 2010](#)). Furthermore, the  $R^2$  is indicative of the variable variance clarification; a value of .02 is rated as low, a value of .13 moderate, and a value of .26 as high ([Cohen, 1988](#)). The indirect effect between variables was considered significant if the confidence interval did not include zero ([Zhao, Lynch, & Chen, 2010](#)).

## 3. Results

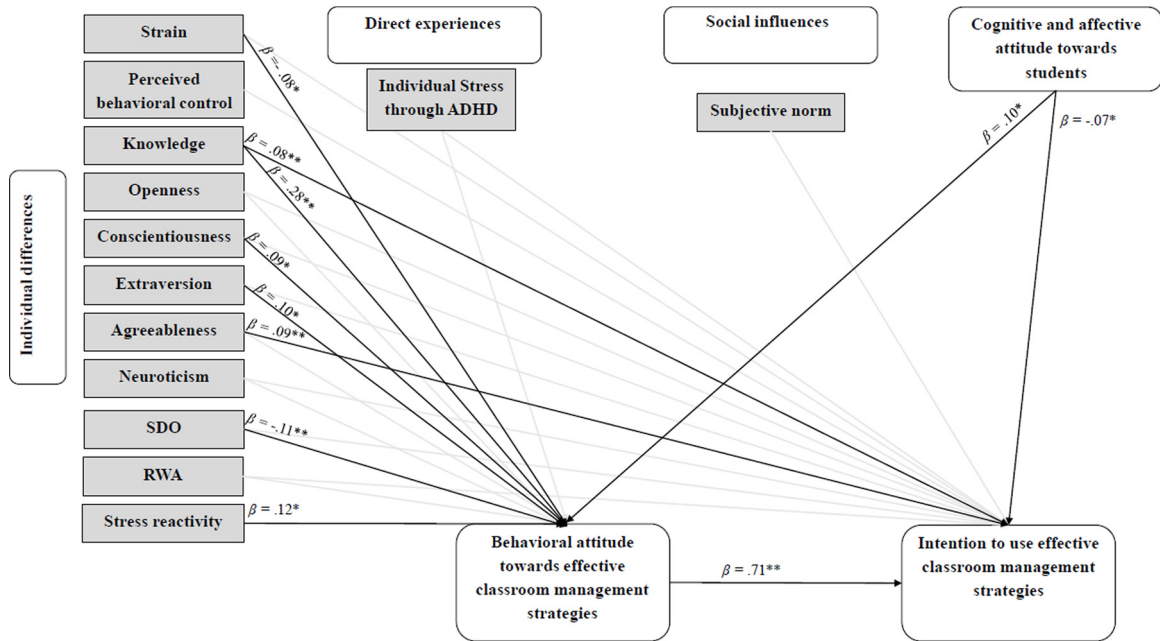
In answering our research questions regarding which variables have the most influence on pre-service teachers’ intention to use CMS, the following calculations were carried out. The correlations between the variables gave preliminary indications of what was relevant, explaining the intention of pre-service teacher to use CMS. All inter-correlations are shown in [Table 2](#).

Two path analyses were run. To answer the first and the second research question, a path analysis was carried out that focused on

Table 2

Inter-correlation between the variables. Notes: \*\* < .01; \* < .05; O = Openness, C = Conscientiousness, E = Extraversion, A = Agreeableness, N = Neuroticism, SDO = Social dominance orientation, RWA = Right-wing authoritarianism.

|                                      | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10     | 11     | 12      | 13      |
|--------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|--------|--------|---------|---------|
| 1. Knowledge                         | 1       |         |         |         |         |         |         |         |         |        |        |         |         |
| 2. Attitude towards CMS              | -.088** | 1       |         |         |         |         |         |         |         |        |        |         |         |
| 3. Attitude towards effective CMS    | .310**  | .766**  | 1       |         |         |         |         |         |         |        |        |         |         |
| 4. Attitude towards ineffective CMS  | -.249** | .582**  | -.076*  | 1       |         |         |         |         |         |        |        |         |         |
| 5. Attitude towards students         | -.092** | .085**  | .098**  | .006    | 1       |         |         |         |         |        |        |         |         |
| 6. Perceived control                 | .041    | .107**  | .112**  | .025    | .241**  | 1       |         |         |         |        |        |         |         |
| 7. Subjective norm - shortened       | .071*   | .070*   | .099**  | -.022   | -.123** | -.204** | 1       |         |         |        |        |         |         |
| 8. Stress react.                     | .037    | .031    | .050    | -.048   | -.184** | -.193** | .321**  | 1       |         |        |        |         |         |
| 9. Strain (BSI)                      | -.137** | -.017   | -.111** | .112**  | -.012   | -.076*  | .043    | .344**  | 1       |        |        |         |         |
| 10. Stress (self - report)           | .017    | .006    | -.018   | .031    | -.341** | -.314** | -.206** | .167**  | .029    | 1      |        |         |         |
| 11. Intention to use CMS             | .116    | .573**  | .437**  | .336**  | -.183** | -.041   | .149**  | .115**  | -.015   | .214** | 1      |         |         |
| 12. Intention to use effective CMS   | .316**  | .531**  | .720**  | -.088** | -.031   | .081**  | .126**  | .109**  | -.075*  | .058   | .694** | 1       |         |
| 13. Intention to use ineffective CMS | -.179** | .216**  | -.178** | .560**  | -.223** | -.146** | .072    | .041    | .064    | .229** | .626** | -.128** | 1       |
| SDO                                  | -.149** | -.079** | -.218** | .161**  | -.087** | -.106** | -.070*  | -.027   | .038    | .089*  | -.033  | -.191** | .167**  |
| RWA                                  | -.171** | -.010   | -.189** | .223**  | -.203** | -.030   | -.019   | .035    | -.072*  | .172** | .088** | -.131** | .258**  |
| O.                                   | .042    | .068*   | .121**  | -.048   | .074*   | .062*   | -.003   | -.072*  | -.023   | -.061* | .007   | .109**  | -.108** |
| C.                                   | .076*   | .126*   | .178**  | -.032   | -.010   | .068*   | .051    | -.063*  | -.090** | .005   | .048   | .129**  | -.075*  |
| E.                                   | .098**  | .136**  | .151**  | .017    | .024    | .193**  | -.064*  | -.238** | -.185** | -.043  | .079** | .132**  | -.034   |
| A.                                   | .109**  | .155**  | .210**  | -.039   | .051    | .071*   | .103**  | -.058   | -.114** | -.016  | .107** | .213**  | -.087** |
| N.                                   | -.026   | -.031   | -.021   | -.018   | -.065*  | -.159** | .238**  | .626**  | .353**  | .096** | .068*  | .034    | .056    |



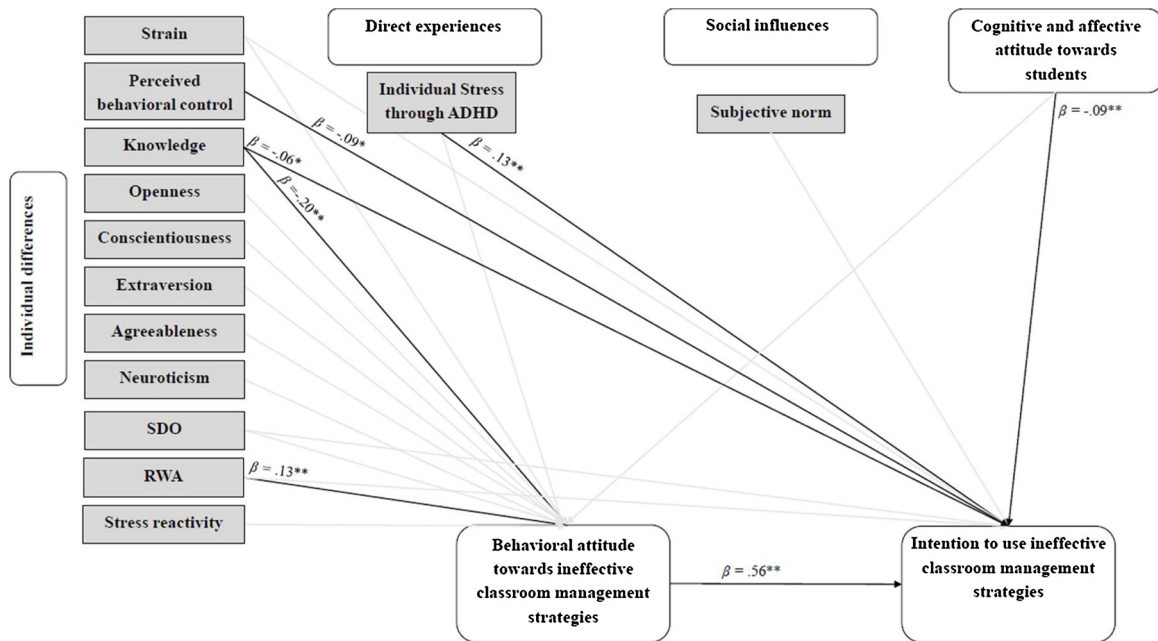
**Fig. 2.** Path model showing the variables that influence the intention to use effective CMS. Grey arrows are included variables in the analysis, but do not significantly explain variance. Black arrows represent significant relevant variables (\* < .05, \*\* < .01). Path models' fit Indices:  $R^2_{intention} = .59^{**}$ ;  $R^2_{behav. attitude} = .20^{**}$ ; test of model fit:  $\chi^2(3) = 6.66$ ,  $p = .08$ ; test for the baseline model:  $\chi^2(29) = 833.08$ ,  $p < .01$ ; CFI = 1.00; TLI = .96; RMSEA = 0.04; SRMR = 0.01.

effective CMS, in which intention to use and attitude towards effective CMS were the dependent variables. Fig. 2 shows the outcomes and calculated fit indices.

The path model explaining the intention to use effective CMS had a multiple determination coefficient of  $R^2 = .59$ , implying that 59 % of the variance is clarified, indicative of a high variance clarification (Cohen, 1988). Positively influencing variables regarding intention were behavioral attitude towards effective CMS ( $\beta = .71$ ,  $p < .01$ ), knowledge ( $\beta = .08$ ,  $p < .01$ ), and agreeableness ( $\beta = .09$ ,  $p < .01$ ); attitude towards the students with ADHD ( $\beta = -.07$ ,  $p < .05$ ) was a negatively influencing variable. The variance clarification of the behavioral attitude towards effective CMS was at 20 % ( $R^2 = .20$ ), which can be defined as a moderate to high clarification, according to Cohen (1988). Influencing variables were knowledge ( $\beta = .28$ ,  $p < .01$ ), conscientiousness ( $\beta = .09$ ,  $p < .01$ ), extraversion ( $\beta = .10$ ,  $p < .05$ ), SDO ( $\beta = -.11$ ,  $p < .01$ ), stress reactivity ( $\beta = .12$ ,  $p < .05$ ), strain ( $\beta = -.08$ ,  $p < .05$ ), and attitude towards students with ADHD ( $\beta = .10$ ,  $p < .01$ ). All mediations with the intention to use effective CMS as the dependent variable and behavioral attitude towards effective CMS as intermitting variable were significant as their 95 %-confidence interval did not include zero: knowledge ( $\beta = .20$  [95 %-CI: 0.15, 0.25]), conscientiousness ( $\beta = .07$  [95 %-CI: 0.18, 0.12]), extraversion ( $\beta = .07$  [95 %-CI: 0.01, 0.12]), SDO ( $\beta = -.08$  [95 %-CI: - 0.14, - 0.03]), strain ( $\beta = -.06$  [95 %-CI: - 0.11, - 0.01]), attitude towards students with ADHD ( $\beta = .07$  [95 %-CI: 0.03, 0.13]), and stress reactivity ( $\beta = .08$  [95 %-CI: 0.02, 0.15]). Fit indices for the whole model all showed a high model fit (test of the model fit:  $\chi^2 [3] = 6.66$ ,  $p = .08$ ; test for the baseline model:  $\chi^2 [29] = 833.08$ ,  $p < .01$ ; CFI = 1.00; TLI = .96; RMSEA = 0.04; SRMR = 0.01).

To answer the third and the fourth research question, a path analysis was carried out that focused on ineffective CMS, in which intention to use and attitude towards effective CMS were the dependent variables. Fig. 3 shows the outcomes and calculated fit indices.

The path model focusing on ineffective CMS explained 44 % of the intentions' variance. Significantly influencing variables were behavioral attitude towards ineffective CMS ( $\beta = .56$ ,  $p < .01$ ), perceived behavioral control ( $\beta = -.09$ ,  $p < .01$ ), knowledge ( $\beta = -.06$ ,  $p = .03$ ), individual stress related to students' ADHD ( $\beta = .13$ ,  $p < .01$ ), and attitude towards students with ADHD ( $\beta = -.09$ ,  $p < .01$ ). The variance explanation of the behavioral attitude towards ineffective CMS was 10 % ( $R^2 = .10$ ), which was defined as low to moderate. Influencing variables were knowledge ( $\beta = -.20$ ,  $p < .01$ ) and RWA ( $\beta = .13$ ,  $p < .01$ ). All mediations with the intention to use ineffective CMS as the dependent variable and the behavioral attitude towards ineffective interventions as the intermitting variable were significant as their 95 %-confidence interval did not include zero: knowledge ( $\beta = -.11$  [95 %-CI: - 0.12, - 0.01]) and RWA ( $\beta = .07$  [95 %-CI: 0.03, 0.12]). Fit indices for the whole model all showed a high model fit (test of model fit:  $\chi^2 [8] = 10.18$ ,  $p = .25$ , test for the baseline model:  $\chi^2 [29] = 516.21$ ,  $p < .01$ ; CFI = 1.00; TLI = .98; RMSEA = 0.02; SRMR = 0.02).



**Fig. 3.** Path model showing the variables that influence the intention to use ineffective CMS. Grey arrows are included variables in the analysis, but do not significantly explain variance. Black arrows represent significant relevant variables (\* < .05, \*\* < .01). Path models' fit Indices:  $R^2_{intention} = .44^{**}$ ;  $R^2_{behav. attitude} = .10^{**}$ ; test of the model fit:  $\chi^2(8) = 10.18$ ,  $p = .25$ ; test for the baseline model:  $\chi^2(29) = 516.21$ ,  $p < .01$ ;  $CFI = 1.00$ ;  $TLI = .98$ ;  $RMSEA = 0.02$ ;  $SRMR = 0.02$ .

#### 4. Discussion

The aim of this study was to examine why the use of effective CMS is low, even though evidence supporting the effectiveness of CMS is well documented and ineffective CMS are still used. At first, we summarize the results with the aim to answer the research questions and afterwards we discuss the results regarding the variables.

The first research question ("Can the intention to use effective CMS be explained by pre-service teachers' behavioral attitude towards effective CMS as well as through individual differences, direct experiences, social influences, and their attitude towards students with ADHD?") is partly confirmed as the tested model explains a large amount of variance with 59 % of pre-service teachers' intentions to use effective CMS and results in good model fit indices. Mostly the attitude towards the CMS, the attitude towards students with ADHD and individual differences are important, whereas direct experiences and social norms do not have significant explanatory power.

The second research question ("Are all variables that influence behavioral attitude towards effective CMS also influencing the intention to use effective CMS mediated through behavioral attitude towards effective CMS?") is confirmed as every variables that has an explanatory effect regarding the attitude towards effective CMS has additionally a significant indirect effect on the intention to use effective CMS.

The third research question ("Can the intention to use ineffective CMS be explained by pre-service teachers' behavioral attitude towards ineffective CMS as well as through individual differences, direct experiences, social influences, and their attitude towards students with ADHD?") is partly confirmed. The amount of variance explained to use ineffective CMS' is high and the tested model shows good model fit indices. However, it also shows that social influences as opposed to individual differences, attitude variables and direct experiences do not contribute to variance explanation.

The fourth research question ("Are all variables that influence behavioral attitude towards ineffective CMS also influencing the intention to use ineffective CMS mediated through behavioral attitude towards ineffective CMS?") is confirmed as every variables that has an explanatory effect regarding the attitude towards ineffective CMS has additionally a significant indirect effect on the intention to use ineffective CMS.

##### 4.1. Expectations and attitude

Behavioral attitude towards CMS is the most important factor in influencing the intention to use both effective and ineffective CMS. This meets Ajzens' (2005) assumption, that the expectation a behavior will show effective consequences increases the intention to perform that behavior. This means pre-service teachers will intend to use CMS as long as they are expected to be effective, whether they are actually effective or not. Thus, it is more important to believe in the effectiveness of CMS than to simply know about them. This is an effect found not only in our analysis. A Cochrane review regarding helping strangers in emergency situations concluded

that imparting knowledge alone would not lead to increased helping behavior. It was also necessary to overcome assumed obstacles or, in other words, make individuals believe their behavior would be effective (Van de Velde et al., 2009).

Furthermore, attitude towards effective CMS is influenced by pre-service teachers' attitude towards students with ADHD. No similar effect was found for ineffective CMS. This indicates that the better a pre-service teacher's attitude is towards students with ADHD, the easier it will be for the pre-service teacher to believe CMS will be effective, and the stronger his or her intention will be to use them. Various studies have shown that teachers often feel more stress from, and have a more critical attitude towards, students with ADHD (Dort et al., under revision; Ghanizadeh, Bahredar, & Moeini, 2006; Greene et al., 2016). It is necessary to improve (pre-service) teachers' attitudes towards students with ADHD to open a mental door to the use of effective CMS.

A more positive attitude towards students with ADHD decreased pre-service teachers' intention to use both effective and ineffective CMS. A possible interpretation of this finding would be that a generally better opinion of students with ADHD is good in changing attitudes towards CMS, but at the same time, it decreases the subjective pressure to act.

#### 4.2. Direct experiences

Individual stress pre-service teacher expect to experience when teaching students with ADHD played an important role in the intention to use ineffective CMS. This finding indicates that higher stress through students with ADHD leads to the intention to use ineffective CMS. Furthermore, it shows that expectations of stress are more important than currently perceived strain measured through the BSI. Performance under stress usually follows an inverted U-function; the performance increases until a stress-peak is reached, and is then followed by a performance decrease (Anderson, 1976). The same mechanism seems to be relevant for pre-service teachers.

#### 4.3. Social influences

Subjective norms are not relevant in influencing the intention to use CMS or attitudes towards it. Presumably, this is due to the participants we included in our study. Pre-service teachers usually do not work in schools; thus, they do not know, and probably do not care, what other teachers may think about CMS.

#### 4.4. Individual differences

Psychological strain shows a relevant impact on influencing the attitude towards CMS. It has a negative effect towards the attitude of effective CMS. Teachers experience a lot of job stress and dissatisfaction that can lead to a high prevalence of mood disorders (Howard et al., 2017). It is easy to understand why higher psychological strain would lead to decreased attitude towards effective CMS. The more one feels stressed, the less one is engaged in daily job tasks, and the worse is the attitude towards effective CMS. This psychological strain, paired with disruptive students, presents (pre-service) teachers with difficult tasks, and makes reliance on effective CMS less likely.

Perceived behavioral control was only relevant in negatively influencing the intention to use ineffective CMS, and thus seemed to have a preventive function. The higher perceived control, the less ineffective and possible harmful CMS were intended to be used, but perceived control did not increase the usage of effective CMS. Furthermore, pre-service teachers are not used to working in schools, thus they know less about possibilities to use CMS. However, if they feel they can effectively teach students with ADHD, they do not seem to want to use ineffective CMS.

Knowledge increased the intention to use effective CMS as well as improved the attitude towards those strategies, while also decreasing the intention to use ineffective CMS and the attitudes towards them. This influence of knowledge is consistent with findings of previous studies (Bekle, 2004). The effect on attitude towards CMS is much higher than on the intention to use them (Ajzen, Joyce, Sheikh, & Cote, 2011). A knowledge-minimum seems to be a necessary precondition, but knowledge alone is not sufficient to change one's intentions (DiClemente, 1989; Fisher, 1992).

Only agreeableness had an effect on the intention to use effective CMS. Regarding the intention to use ineffective CMS, no personality variable had an influence. This is nearly consistent with the findings of Ajzen (2005), who said that personality changes the attitude, but not the intention behind or the execution of the behavior in question. Agreeableness, according to DeYoung (2015), leads to cooperative and altruistic behavior, and is able to help explain why some pre-service teachers, despite challenges, intended to use effective CMS.

Conscientiousness, extraversion, stress reactivity positively, and SDO negatively influenced pre-service teachers' attitudes towards effective CMS, while attitudes towards ineffective CMS were positively influenced by RWA. Effective CMS are often strategies that do not emphasize hierarchical disparities between (pre-service) teachers and students and aim to downsize hierarchies. For pre-service teachers who rate high on RWA and SDO, it would be important to focus not only on the effectiveness of CMS, but to emphasize that the benefits of CMS do not challenge their authority. Conscientiousness and extraversion seemed to be helpful in generating a more positive attitude towards the use of effective CMS, and explains why the attitude varies widely. DeYoung (2015) offers explanations for this phenomenon. More conscientious pre-service teachers tend to prioritize non-immediate goals (DeYoung, 2015), and this could help them to be more open towards strategies that might need some time to show their full effect. More extraverted pre-service teachers tend to be more goal-oriented (DeYoung, 2015), and most likely feel the use of effective CMS promises to be more worthwhile.

#### 4.5. Implications

Taken together, primarily the attitude towards CMS influences the intention to use more effective and fewer ineffective CMS. Individual differences influence intention as well, but to varying degrees. Direct experiences as well as individual differences not only affect the attitude, but also the intention to use CMS. Thus, for future training with teachers, a focus on those expectations and therefore also the attitude is highly relevant. Pre-service teachers not only need more knowledge about CMS, but also the conviction that effective CMS are efficacious. Furthermore, psychological strain reduces the intention to use effective CMS indirectly, and perceived individual stress through ADHD increases the intention to use ineffective CMS, which suggests that even pre-service teachers have to learn how to deal with stress in their everyday live. Perceived behavioral control seems to have a preventive function regarding ineffective CMS, implicating that the more they know what they are capable of the less they will rely on ineffective CMS. Teacher training should thus focus on increasing the conviction to be able to teach students with ADHD, that the known CMS will work and that there are effective strategies to cope with stress and strain.

How those expectations and therefore also the attitude can be violated and shaped, and what kind of coping strategies are obstacles to this shaping process, should be clarified through future research. For example, it could be that (pre-service) teachers tend to immunize when an effective CMS works with a student with ADHD and then they will not use it again (e.g. “this worked once, but that is an exception”). Conversely, this can happen when using an ineffective CMS. Even if using an ineffective CMS does not improve student behavior, immunization can occur and the ineffective strategy is used again. The mechanism found in this study should be tested for relevance in other subgroups and experimentally. Therefore, a study with current teachers is already completed, and a report of its findings will follow. An analysis that includes different types of schools may be relevant.

#### 4.6. Limitations

The findings of the present study allow us to inspect the nature of attitude and give us preliminary evidence for what may be important for in-service teachers. Additionally, the results illustrate that we should not solely focus on pre-service teachers, as they have no experience putting CMS into practice.

The TPB is based on the assumption that rational choices underlie behavior, and some researchers argue that human behavior is not rational (Fazio, 1990). Teachers' behavior in the classroom is, in most cases, prepared and should be viewed as rational; however, behavior is, of course, not rational in all cases. Even if the TPB is not the perfect picture of reality, it is at least good enough to predict behavior properly.

Another issue that we have to face is the *intention-behavior gap* implicating the fact that a medium to high change in intention just leads to a small to medium change in the actual behavior (Sheeran & Webb, 2016). As we only measure the intention to use CMS the actual behavior could be a bit different from the intention, though intention is still the best predictor for the actual behavior (Sheeran & Webb, 2016). In experimental surveys this issue needs to be tested and further possibilities to overcome this gap, like making the behavior easier to perform or make the goal part of their identity need to be considered (Sheeran & Webb, 2016). At the moment, we are planning a virtual reality study with which we aim to investigate how teachers behave in a virtual classroom when a student shows symptoms of ADHD.

The direct measure of attitude is disputed for good reasons. It can be assumed that there are differences between attitudes measured implicitly and explicitly, and both need to be assessed (Ajzen, 2005; Niemand & Fleischhauer, 2012). The cross-sectional design was also a limitation, as it did not allow us to make more than correlative assumptions about effects or the impact of variables. Furthermore, the BFI is not very much focused on details and another measurement for personality variables would probably have provided more detailed results. However, we decided to use the BFI to keep the online survey as short as possible, to avoid an overload of questions and ensure a high participation rate.

Due to SoSciSurveys' data privacy limitations it was not possible to get more information about the dropped out pre-service teachers. Therefore, it is not possible to compare the dropped out with the pre-service teachers that completed the survey.

#### 4.7. Conclusion

The first examination of the presented path model suggests for effective CMS to be put into practice, it is necessary that pre-service teachers expect effective CMS to be effective and ineffective CMS should be classified as ineffective. Furthermore, direct experiences and individual differences have a shaping function regarding the attitude towards CMS and thus also towards expectations. The attitude towards CMS needs to be changed to increase the probability pre-service teachers will integrate CMS into their daily routines. We are currently conducting a virtual reality investigation to examine those findings experimentally.

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